

# EECE 306 Lab 5 notebook. Flux, Gauss's Law, and the Divergence

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Fill in every TODO, run `publish('Lab05_notebook.m')` from the toolbox root, print the HTML to PDF, three to five pages.

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## Setup

---

```
clear; close all;  
Q = 3e-9;
```

## Gauss's law on the sphere

---

```
s = em.src.pointCharge(Q, [0 0 0]);  
FD = @(r) em.field.D(s, r);  
sphR = @(th,ph) 0.5*[sin(th)*cos(ph) sin(th)*sin(ph) cos(th)];  
Phi = em.field.flux(FD, sphR, [0 pi], [0 2*pi], 60, 120);  
fprintf('flux through sphere = %.6e C\n', Phi);  
fprintf('enclosed charge = %.6e C\n', Q);  
fprintf('relative error = %.3e\n', abs(Phi - Q)/Q);
```

```
flux through sphere = 3.000343e-09 C  
enclosed charge = 3.000000e-09 C  
relative error = 1.142e-04
```

## The same charge outside the surface

---

```
sout = em.src.pointCharge(Q, [5 0 0]);  
Phi0 = em.field.flux(@(r) em.field.D(sout, r), sphR, [0 pi], [0 2*pi], 60, 120);  
fprintf('flux with charge outside = %.3e C (consistent with zero)\n', Phi0);  
% TODO justify the absolute tolerance you would use to call this zero.  
% Compare it against the flux each hemisphere carries separately.  
% The x-positive and x-negative hemispheres carry about -7.444e-12 C
```

```
% and +7.445e-12 C. Their cancellation makes relative error against zero
% undefined. For the starter midpoint grid we use 4e-16 C (about 5e-5 of
% a hemisphere flux); its residual is 3.379e-16 C. The refined Gauss test
% uses 8e-18 C, about 1e-6 of a hemisphere flux, and checks refinement.
PhiRight = em.field.flux(@(r) em.field.D(sout,r), sphR, ...
    [0 pi], [-pi/2 pi/2], 60, 120);
PhiLeft = em.field.flux(@(r) em.field.D(sout,r), sphR, ...
    [0 pi], [pi/2 3*pi/2], 60, 120);
fprintf('right / left hemisphere = %.6e / %.6e C\n', PhiRight, PhiLeft);
```

```
flux with charge outside = 3.379e-16 C (consistent with zero)
right / left hemisphere = -7.444250e-12 / 7.444588e-12 C
```

### Three surfaces, one answer

The same charge, placed off center inside all three closed surfaces.

```
s3 = em.src.pointCharge(Q, [0 0.05 0.1]);
FD3 = @(r) em.field.D(s3, r);
PhiSph = em.field.flux(FD3, sphR, [0 pi], [0 2*pi], 60, 120);
cube = 0;
faces = { @(u,v)[u v 0.5*ones(size(u))], @(u,v)[v u -0.5*ones(size(u))], ...
    @(u,v)[v 0.5*ones(size(u)) u], @(u,v)[u -0.5*ones(size(u)) v], ...
    @(u,v)[ 0.5*ones(size(u)) u v], @(u,v)[-0.5*ones(size(u)) v u] };
for k = 1:6
    cube = cube + em.field.flux(FD3, faces{k}, [-0.5 0.5], [-0.5 0.5], 60, 60);
end
hemi = @(th,ph) 0.5*[sin(th)*cos(ph) sin(th)*sin(ph) cos(th)];
capD = @(u,v) [u.*cos(v) -u.*sin(v) zeros(size(u))];
PhiHemi = em.field.flux(FD3, hemi, [0 pi/2], [0 2*pi], 60, 120) ...
    + em.field.flux(FD3, capD, [0 0.5], [0 2*pi], 60, 120);
fprintf('sphere          %.6e C\n', PhiSph);
fprintf('cube            %.6e C\n', cube);
fprintf('hemisphere and cap %.6e C\n', PhiHemi);
% TODO which surface was hardest to set up correctly and why. Face
% orientation is the usual culprit, state how each outward normal was
% checked.
% The cube was hardest because six independent parameterizations must
% agree on outward orientation. In the listed face order, ru cross rv
% gives +z, -z, +y, -y, +x, -x; constant Cartesian fields independently
% verified each signed unit area. On the sphere the cross product points
% radially outward, while capD gives -u*z, downward out of the upper
% hemisphere. A constant +z field integrates to -pi*(0.5)^2 on that cap.
```

```
sphere          3.000379e-09 C
cube            3.000204e-09 C
hemisphere and cap 3.000398e-09 C
```

### Independence of the charge position inside

```
for x0 = [0 0.15 0.3]
    sm = em.src.pointCharge(Q, [x0 0.1 0]);
```

```

Pm = em.field.flux(@(r) em.field.D(sm, r), sphR, [0 pi], [0 2*pi], 60, 120);
fprintf('charge at x = %.2f    flux = %.6e C\n', x0, Pm);
end

```

```

charge at x = 0.00    flux = 3.000323e-09 C
charge at x = 0.15    flux = 3.000285e-09 C
charge at x = 0.30    flux = 3.000207e-09 C

```

## Divergence against the density, an element count study

A uniform ball of charge. The derivative step  $h$  must average over many elements, so  $h = 0.05$  m is passed explicitly, and the element count is raised at fixed  $h$ .

```

rhov = 1e-6;
hdiv = 0.05;
for Nu = [10 20 40]
    ball = @(u,v,w)[w*sin(u)*cos(v) w*sin(u)*sin(v) w*cos(u)];
    sb = em.src.volCharge(rhov, ball, [0 pi], [0 2*pi], [0 0.5], Nu, 2*Nu, Nu);
    dv = em.op.div(@(r) em.field.D(sb, r), [0.1 0.05 0.1], hdiv);
    fprintf('Nu = %2d    div D inside = %.4e    rho_v = %.4e    rel err = %.2e\n', ...
        Nu, dv, rhov, abs(dv - rhov)/rhov);
end
dfree = em.op.div(FD, [2 1 1], hdiv);
fprintf('div D in charge free space = %.3e (consistent with zero)\n', dfree);
% TODO one or two sentences. Why must h span many source elements here,
% and what would div D return with h far smaller than the element spacing.
% A step spanning many source elements samples the smooth volume field;
% a much smaller step resolves charge-free gaps between discrete point
% charges and returns nearly zero divergence away from those charges.
% Here 2,000 elements give 118% error, 16,000 give 0.639%, and 128,000
% give 0.398% at the starter point; 128,000 is our reported checked count.
% At three interior points the tests give at most 0.953% error at 128,000
% and 1.715% at 1,024,000 elements, with less than 3% change on refinement.
% At the starter point and 1,024,000 elements, h=1e-5 m gives only
% 8.48e-12 C/m^3, not the continuum density of 1e-6 C/m^3.

```

```

Nu = 10    div D inside = 2.1807e-06    rho_v = 1.0000e-06    rel err = 1.18e+00
Nu = 20    div D inside = 9.9361e-07    rho_v = 1.0000e-06    rel err = 6.39e-03
Nu = 40    div D inside = 1.0040e-06    rho_v = 1.0000e-06    rel err = 3.98e-03
div D in charge free space = 1.184e-14 (consistent with zero)

```

## Orientation is a convention you must own

```

sphFlip = @(ph,th) 0.5*[sin(th)*cos(ph) sin(th)*sin(ph) cos(th)];
PhiFlip = em.field.flux(FD, sphFlip, [0 2*pi], [0 pi], 120, 60);
fprintf('flux with swapped parameter order = %.6e C (sign flipped)\n', PhiFlip);

```

```

flux with swapped parameter order = -3.000343e-09 C (sign flipped)

```

## Interpretation

TODO three to six sentences. Gauss's law was verified without knowing any closed form for the flux integral. Explain why this class of test remains available on a problem with no known answer. Gauss's law compares an integrated computed field with the independently known enclosed charge, so we do not need a closed form for the field. Agreement for a sphere, a cube, and a closed hemisphere tests geometry as well as conservation. The merged-source test gives  $5e-9$  C from the three inside charges, even with two outside charges present. This test remains available for irregular sources and surfaces, but quadrature refinement and independent geometry checks are still needed because conservation alone cannot rule out every local field error.

## Problems encountered

TODO honest account, or NONE. The preserved 60-by-120 midpoint sphere grid has  $1.142e-4$  relative error, above the handout's  $1e-6$  target; tests meet that target with Gauss quadrature and also with a 960-by-1920 midpoint grid ( $4.46e-7$ ). The handout says to use the default  $h=1e-5$  m, but the starter explicitly sets  $h=0.05$  m for volume elements; we preserved it and tested both. Refinement is not uniform: at  $[0.04 - 0.1 \ 0.15]$  m, 1,024,000 elements still give 15.3% divergence error at  $h=0.05$  m, versus 3.18% at  $h=0.1$  m. The reported five-percent result therefore applies to the tested points and step, rather than establishing accuracy everywhere in the ball.

## Full test suite

```
runTests
```

```
EECE 306 toolbox test suite
```

```
-----
```

```
eta =
```

```
376.7303
```

```
test_lab01 checks complete
```

```
PASS test_lab01          0.19 s
```

```
test_lab02 checks complete
```

```
PASS test_lab02          0.33 s
```

```
5000-point field evaluation: 0.0011 s
```

```
test_lab03 checks complete
```

```
PASS test_lab03          0.10 s
```

```
test_lab04 checks complete
```

```
PASS test_lab04          0.10 s
```

```
merged source: flux 5.00000000e-09 C, enclosed 5.00000000e-09 C
```

```
density errors at 128000 / 1024000 elements: 0.953% / 1.715% max
```

```
sphere / cube / closed hemisphere: 3.00000000e-09 / 3.00000000e-09 / 3.00000000e-09 C
```

```
sphere Gauss rel err 1.42e-11; midpoint N=960 rel err 4.46e-07
```

```
test_lab05 checks complete
```

```
PASS test_lab05          1.90 s
```

```
-----
```

```
5 passed, 0 failed, 5 total
```

```
All tests passing.
```